The Ground Beneath the Ought

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Quantum Coherence Ethics: A Pre-Ethical Framework

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Light strikes the prism.
Nothing was added.

For Hugh Everett III —

for every branch of him that never heard he was right.

And for everyone who came before him —

who half created and half perceived

the ground they were standing on,

without the physics to see it whole.

Contents

Chapter 1

The Question

There is no longer any independent system state or observer state,
although the two have become correlated in a one-one manner.

— Hugh Everett III, *Reviews of Modern Physics* 29 (1957)

In 1957, Hugh Everett III proposed that the wave function never collapses — that every quantum event that can occur does occur, each in its own branch of a continuously branching reality, all branches equally real. The proposal was largely ignored. The physics establishment, anchored to Bohr's Copenhagen interpretation, found it unnecessary. Everett left physics, took a job as a defense analyst, and died at fifty-one in 1982 without seeing his idea taken seriously.

The wave function does not collapse. The branches are real. This is now defended rigorously by Deutsch, Carroll, Wallace, and others, supported by the decoherence program of Zurek and collaborators, and taken seriously by a substantial portion of the physics community. The arguments continue — the Born rule derivation under MWI remains the most contested open problem in the foundations of physics — but the interpretation is no longer fringe.[1]

This paper is not about the physics. It is about what the physics means — specifically, what it means for ethics, for the structure of what a person actually

[1] Hugh Everett III, "'Relative State' Formulation of Quantum Mechanics," *Reviews of Modern Physics* 29, no. 3 (1957): 454–462. For subsequent development see David Deutsch, *The Fabric of Reality* (London: Allen Lane, 1997); Sean Carroll, *Something Deeply Hidden* (New York: Dutton, 2019); David Wallace, *The Emergent Multiverse* (Oxford: Oxford University Press, 2012).

is, for the question of how to live. It asks: if the branches are real and every downstream self is real, what does the present self owe them?

That question has a structural answer. This paper presents it.

The argument is conditional throughout: *if* the Many Worlds Interpretation is correct, *then* what follows. Readers who dispute MWI should engage the physics literature directly. The paper makes no claim beyond the conditional. But the conditional carries more weight than it might appear — because if Everett is right, then every prior ethical framework was working with an incomplete description of what a person is. And incomplete descriptions produce incomplete ethics, however carefully reasoned.

Chapter 2

Standing on Shoulders

What Got Me Here

O brave new world, that has such people in't.

— William Shakespeare, *The Tempest*, V.i

PHILOSOPHY is not done in isolation. It is done in conversation with the dead, with books read at the wrong age and the right one, with ideas encountered in places you did not expect. The framework presented in this paper would not exist without a specific set of thinkers who, from independent directions and across centuries, kept alive intuitions that become precise only once MWI is taken seriously. They did not point toward this framework. They pointed me toward it. The distinction matters. None of them had access to Everett's physics. All of them shaped the questions I eventually learned to ask.

2.1 Giordano Bruno

GIORDANO Bruno was burned alive in Rome in 1600 for arguing, among other things, that the universe contains infinite worlds. He was doing theology and metaphysics, not physics, and his arguments were philosophical rather than mathematical. But the picture he arrived at — an infinite, unbounded reality containing countless worlds each as real as any other — is structurally identical to the picture Everett would derive from the Schrödinger equation three and a half centuries later. Bruno arrived at it from God. Everett arrived at it from mathematics. Bruno died for it.[1]

The courage of that picture — the willingness to say that what we experience as one world is one instance of a vastly larger structure — was something I needed to see before I could follow the physics where it went. Bruno demonstrated that it is possible to hold the infinite picture and not flinch. That mattered.

2.2 The English Romantic Poets

The English Romantics — Wordsworth, Coleridge, Blake, Keats — were fighting a battle against the closed mechanical universe of Newtonian physics and Paley's watchmaker God that I have always understood as my battle too. They could not articulate what was wrong with the mechanical account in the language the account required, so they articulated it in a different language.

> *. . . a sense sublime*
> *Of something far more deeply interfused,*
> *Whose dwelling is the light of setting suns,*
> *And the round ocean and the living air,*
> *And the blue sky, and in the mind of man:*
> *A motion and a spirit, that impels*
> *All thinking things, all objects of all thought,*
> *And rolls through all things.* — Wordsworth, *Tintern Abbey*, 95–102[2]

Coleridge asked whether all of animated nature might be *"organic Harps diversely framed, / That tremble into thought, as o'er them sweeps / Plastic and vast, one intellectual breeze, / At once the Soul of each, and God of all"*[3] —

[1] Giordano Bruno, *De l'Infinito, Universo e Mondi* (1584), Dialogue V, trans. Dorothea Waley Singer, in *Giordano Bruno: His Life and Thought* (New York: Henry Schuman, 1950), 370.

[2] William Wordsworth, "Lines Written a Few Miles above Tintern Abbey," lines 95–102, in *Lyrical Ballads* (1798), ed. R. L. Brett and A. R. Jones (London: Routledge, 2nd ed., 1991).

[3] Samuel Taylor Coleridge, "The Eolian Harp," lines 44–48 (1817 text), in *The Complete*

reaching toward distributed being in the language of 1795. Blake attacked the Urizenic closed universe directly: *"The bounded is loathed by its possessor. The same dull round, even of a universe, would soon become a mill with complicated wheels."*[4] And Keats described **negative capability** — the capacity to remain *"in uncertainties, Mysteries, doubts, without any irritable reaching after fact and reason"*[5] — the correct epistemic stance toward a branching universe: all possibilities real, none requiring forced resolution.

These poets kept alive in me the conviction that the mechanical account of a person was missing something essential. That conviction is what made me willing to take the distributed being seriously when the physics offered it.

2.3 Aristotle

ARISTOTLE gave ethics its first honest structural question: not what should you do, but what must be true for human existence to go well at all? His answer — *eudaimonia*, the activity of the soul in accordance with virtue — is the original Triple P intuition. *"Every art and every inquiry, and similarly every action and pursuit, is thought to aim at some good; and for this reason the good has rightly been declared to be that at which all things aim."*[6]

What I saw in Aristotle was not just the precondition of ethics but the shape of what QCE would later formalize. If you set up every future branch-self with maximum probability, possibility, and potential — if you invest in the conditions that allow every downstream self to flourish fully — what you get is not a single good life but an exponentially compounding distribution of flourishing. Human flourishing under QCE is not linear. It is multiplicative. Every investment in

Poetical Works of Samuel Taylor Coleridge, ed. Ernest Hartley Coleridge (Oxford: Oxford University Press, 1912), vol. 1, 101.

[4] William Blake, *The Marriage of Heaven and Hell*, Plate 14, in *The Complete Poetry and Prose of William Blake*, ed. David V. Erdman (New York: Anchor Books, rev. ed., 1988), 40.

[5] John Keats, letter to George and Thomas Keats, December 21–27, 1817, in *The Letters of John Keats*, ed. Hyder Edward Rollins (Cambridge: Harvard University Press, 1958), vol. 1, 193.

[6] Aristotle, *Nicomachean Ethics*, 1.1, 1094a1–3, trans. W. D. Ross, rev. J. L. Ackrill and J. O. Urmson (Oxford: Oxford University Press, 1998).

Triple P propagates forward through branching and compounds. Aristotle was pointing at the structure. He just did not have the branching to show him how deep it goes.

2.4 Epicurus

Epicurus named the root problem. Fear of death, he argued, is the primary distortion of human life — the anxiety that drives people to accumulate, to dominate, to sacrifice the present for a future that may never come. Every Newtonian ethical framework is, in some sense, a sophisticated response to mortality anxiety. Epicurus saw the problem with perfect clarity even if his solution was incomplete. QCE does not answer the death problem. It dissolves it, by correcting the description of what dies.

2.5 Marcus Aurelius

Marcus Aurelius gave me Principle 2 in 170 CE without knowing it: *"What injures the hive injures the bee."*[7] The self as a node in a larger whole, owing duties to that whole not from sentiment but from structural participation in it. The Stoic *amor fati* — the love of what is, the acceptance of the full structure of existence rather than fighting for one narrow outcome — is proto-Non-Tyranny-of-the-Present. He was working with the wrong metaphysics. The emotional and ethical ground he was standing on is real.

[7] Marcus Aurelius, *Meditations*, VI.54, trans. Gregory Hays (New York: Modern Library, 2002).

2.6 Thomas Hobbes

THOMAS Hobbes looked at the same problem Rand would later examine — how to get ethical behavior out of rational self-interested agents — and gave the most honest Newtonian answer available: you cannot. Without the Leviathan, without an external sovereign enforcing the social contract, rational self-interest produces the war of all against all. Hobbes was not wrong about the Newtonian universe. He was right. The structural solution to his problem does not come from better moral argument. It comes from better physics. Under MWI, the Leviathan is unnecessary because the consequences of defection are already distributed across the agent's own future. You cannot escape your distribution.

2.7 Baruch Spinoza

SPINOZA gave me the courage to turn the ontology. In the *Ethics*, he proposed that there is only one substance — God, or Nature, *Deus sive Natura* — and followed the consequences wherever they led, regardless of what it cost him.

> *A free man thinks of death least of all things; and his wisdom is a meditation not of death but of life.* — Spinoza, *Ethics*, IV.67[8]

That move — taking the ontology seriously, turning it, and following it — is the move QCE learned from Spinoza. The content is different. The method is his.

2.8 Ayn Rand

AYN Rand was among the few ethical thinkers who refused to moralize rational self-interest away — who looked at the Newto-

8. Benedict de Spinoza, *Ethics*, Part IV, Proposition 67, trans. Edwin Curley, in *The Collected Works of Spinoza*, vol. 1 (Princeton: Princeton University Press, 1985).

nian individual and said honestly that ethics must start from that individual's actual interests or it starts from a lie. Her definition of ethics as a structural question before a prescriptive one:

> *The first question is not: What particular code of values should man accept? The first question is: Does man need values at all — and why?* — Rand, *The Virtue of Selfishness*[9]

That is the pre-ethical question. It is the same question QCE asks. Her answer was built for a single timeline and a single future self. QCE does not complete what Rand was building — it replaces the engine while keeping the question. Her engine was volitional consciousness; Foundation 3 removes that requirement entirely. What survives the replacement is her question, which she was braver than almost anyone about asking.

2.9 David Hume

Hume held the boundary firm and would not let me cross it. The is/ought gap is real and correctly identified.[10] QCE does not cross it. But Hume also gave me the bundle theory of the self: *"I never can catch myself at any time without a perception, and never can observe any thing but the perception."*[11] The first honest acknowledgment in Western philosophy that the unified enduring self is not given but constructed. He found this troubling. It is clarifying.

2.10 Immanuel Kant

Kant reached for the structural foundation with both hands and could not quite grasp it. The categorical imperative is the most rigorous attempt in Western philosophy to ground ethics

[9] Ayn Rand, "The Objectivist Ethics," in *The Virtue of Selfishness* (New York: New American Library, 1964), 14.

[10] David Hume, *A Treatise of Human Nature*, 3.1.1, ed. L. A. Selby-Bigge, rev. P. H. Nidditch (Oxford: Clarendon Press, 1978).

[11] Hume, *Treatise*, 1.4.6.

without appealing to consequences or sentiment. He knew he needed freedom and could not prove it, so he postulated it.[12] That honesty is what I respect most about Kant. Foundation 3 of QCE removes the need for the postulation. Ethical weight arises not from freedom of choice but from the structural fact that present actions shape the statistical character of the future distribution. He was building for the right destination. The physics he needed arrived a hundred and fifty years after he died.

2.11 G.W.F. Hegel

HEGEL belongs here, though his contribution may belong more fully to the next paper. The *Aufhebung* — sublation, the preservation of thesis and antithesis within the synthesis rather than their cancellation — anticipates something important about the distributed being. When the wave function branches, both directions are preserved. The present self goes both ways. Thesis and antithesis are not resolved into a synthesis that eliminates them. They are both real. That is Everett. That is also Hegel.

2.12 Derek Parfit

DEREK Parfit came closer than anyone. *Reasons and Persons* is the work of someone who looked at the branching cases directly and refused to look away. He got one move away from the distributed being and made the wrong turn. The nature of that wrong turn — and why the correct turn is to let identity branch rather than dissolve it — is the most important philosophical argument in this paper and is addressed fully in Chapter 5.

[12]Immanuel Kant, *Groundwork of the Metaphysics of Morals*, Ak. 4:429, trans. Mary Gregor and Jens Timmermann (Cambridge: Cambridge University Press, 2012).

2.13 Ludwig Wittgenstein

Wittgenstein cleared the ground. The private language argument dissolves the demand for an inner metaphysical identity-fact that Parfit's position implicitly requires. Without Wittgenstein, the Parfit engagement is harder to make precise. He provided the instrument without knowing what it would eventually be used for.

2.14 Robert Charles Wilson

Robert Charles Wilson wrote a science fiction story called *"Divided by Infinity,"* published in 1998, in which a man discovers he cannot die — not because he is immortal but because there is always a branch in which he survives.[13] The story treats this as horror. Reading it, I had a different response: *so what does that mean for how to live?* That question, asked in an unremarkable afternoon, is where QCE began. Everything else — the formalization, the genealogy, the engagement with Parfit and Wittgenstein and the physics literature — was the attempt to answer it rigorously.

I am grateful to all of them. None of them pointed toward this framework. All of them pointed me.

[13] Robert Charles Wilson, "Divided by Infinity," in *Starlight 2*, ed. Patrick Nielsen Hayden (New York: Tor Books, 1998).

Chapter 3

The Conditional

And what if all of animated nature
Be but organic Harps diversely framed,
That tremble into thought, as o'er them sweeps
Plastic and vast, one intellectual breeze,
At once the Soul of each, and God of all?

— S. T. Coleridge, *The Eolian Harp* (1817)

THE entire argument rests on one premise: the Many Worlds Interpretation of quantum mechanics is correct.

If Hugh Everett III was right — if the wave function never collapses, if the branches are real, if every quantum event that can occur does occur in its own branch — then a person is not what any prior ethical framework assumed them to be. A person is a **distributed being**: a single entity instantiated across every branch of the wave function sharing its causal origin. Not a metaphor. The direct ontological consequence of taking Everett's formulation seriously and following it where it goes.

From inside any branch, it looks like Copenhagen — one outcome, one world, one self. That experience is real and not an illusion. It is the correct description of what experience is from inside a single branch. Foundation 4 of the formal apparatus states it directly: it looks like Copenhagen from the inside. It is Everett from the outside. Both are true.

The prism image carries this precisely. Light carries its wavelengths in superposition — not as potential but as structure. The prism is the unitary evolution of the wave function: the process that takes superposition and makes structure legible, not by collapsing it but by letting every branch become fully itself. What emerges is not what the light became. It is what the light always was, now legible.

> *You are the light.*
> *The branches are already in you.*
> *The prism is the unitary evolution of the wave function —*
> *the process that takes superposition and makes structure legible,*
> *not by collapsing it, but by letting every branch become fully itself.*

From this ontological claim, a structural account of ethics follows. The present self is the ancestor of all downstream branch-selves. A present action does not select an outcome. It shapes the inherited conditions — the resources, the open paths, the starting position — that every downstream self receives. The ethical question is not which branch to choose. It is what to leave for those who follow.

This is **Quantum Coherence Ethics**. The formal apparatus is presented in Chapter 4. The philosophical defense occupies Chapters 5 through 6. The paper closes with two structural corrections to the Newtonian picture of the person that follow from the distributed-being ontology.

Chapter 4

The Formal Apparatus

By reality and perfection I mean the same thing.

— Baruch Spinoza, *Ethics*, Part II, Definition 6

4.1 Foundations

Foundation 1 (Distributed Being)**.** A person is a distributed being — a single entity instantiated across every branch of the wave function sharing its point of causal origin (DNA, spacetime coordinates, causal insertion).

Foundation 2 (Spectral Identity)**.** The distributed being persists across branches in the way light persists across the wavelengths of its spectrum: every wavelength is the light. No branch-self is more fully the distributed being than any other.

Foundation 3 (Independence from Free Will)**.** Ethical weight arises not from freedom of choice — which is not required — but from the structural fact that present actions shape the statistical character of the future distribution.

Foundation 4 (Dual Perspective)**.** It looks like Copenhagen from the inside. It is Everett from the outside.

Foundation 5 (Ancestral Structure)**.** The present self is the ancestor of all downstream branch-selves. A present action does not select an outcome; it shapes the inherited conditions — resources, open paths, starting position — that every downstream self receives. The ethical question is not which branch to choose, but what to leave for those who follow.

4.2 Triple P

Definition 4.1 (Triple P)**.** For any future branch ω, three quantities characterize the quality of existence within it:

Symbol	Name	Measures
$P(\omega)$	Probability	Survival — whether you are alive in this branch
$\Sigma(\omega)$	Possibility	Scope — degrees of freedom, open paths, ways existence can be lived
$\Lambda(\omega)$	Potential	Scale — capacity, resources, health, knowledge, power to act

All quantities non-negative: $P \in [0, 1]$; $\Sigma, \Lambda \geq 0$.

The quality of existence in branch ω is:

$$Q(\omega) = P(\omega) \cdot \left[\Sigma(\omega) + \Lambda(\omega)\right] \tag{4.1}$$

Probability gates both possibility and potential: without survival, neither scope nor capacity has a subject.

Definition 4.2 (Branch Weight)**.** Each future branch ω carries a branch weight $w(\omega \mid a)$: the share of the wave function's amplitude flowing into ω given present action a. This is the Born measure — an ontological quantity describing the structure of reality.

Remark. The branch weight $w(\omega \mid a)$ and the survival probability $P(\omega)$ are both probabilities but different substances. $w(\omega \mid a)$ is *ontological* — how much of reality flows into the branch. $P(\omega)$ is *epistemological* — whether you are alive in it. They must not be conflated.

4.3 The QCE Equation

Let $\Omega(a)$ denote the set of future branches reachable from present action a.

The QCE Equation

$$\mathrm{QCE}(a) = \sum_{\omega \in \Omega(a)} w(\omega \mid a) \cdot P(\omega) \cdot \Big[\Sigma(\omega) + \Lambda(\omega)\Big] \tag{4.2}$$

Remark. The QCE Equation is descriptive. It measures the quality of existence across the downstream distribution resulting from a present action — the inherited conditions available to the distributed being's descendants. It is not a decision procedure. It is not a utility function to be maximized. It is a description of what is at stake.

4.4 Dual Reading

The equation admits two readings corresponding to the inside and outside views of the same reality.

Interpretation 4.1 (Copenhagen — Inside View)**.** The weights are Born probabilities summing to 1. The equation gives expected Triple P from within a single branch:

$$\mathrm{QCE}_C(a) = \mathbb{E}_\omega\Big[Q(\omega)\Big], \qquad \sum_{\omega} w(\omega \mid a) = 1 \tag{4.3}$$

Interpretation 4.2 (Everett — Outside View)**.** The weights are branch measures, unbounded in sum. The equation gives total Triple P across the full distributed being. The multiplicative function of branching lives here.

$$\mathrm{QCE}_E(a) = \sum_{\omega \in \Omega(a)} \mu(\omega \mid a) \cdot Q(\omega), \qquad \sum_{\omega} \mu(\omega \mid a) \text{ unbounded} \tag{4.4}$$

> *Same equation. Different reading of the weight.*
> *Copenhagen from the inside; Everett from the outside.*

4.5 Action Selection

Theoretical (Argmax)

$$a^* = \arg\max_a \ \mathrm{QCE}(a) \tag{4.5}$$

Practical (Softmax)

$$\pi(a) = \frac{\exp\big(\beta \cdot \mathrm{QCE}(a)\big)}{\sum_{a'} \exp\big(\beta \cdot \mathrm{QCE}(a')\big)} \tag{4.6}$$

where $\beta > 0$ is a temperature parameter. As $\beta \to \infty$, softmax collapses to argmax. As $\beta \to 0$, all actions become equiprobable. These formulations demonstrate formal completeness — that the framework can connect to behavior — not that optimization is the prescriptive goal.

4.6 Eight Principles

Principle 1 (Quantum Morality)**.** Good is increasing Triple P across the distribution. Evil is decreasing it.

Principle 2 (The First Rule)**.** What is good for this goose is good for the gander. Harm to others is structural diminishment of one's own distribution: damaged relationships, narrowed trust, reduced scope and scale propagate across every downstream branch the moment the action is taken. Under MWI, rational self-interest converges on cooperation structurally, not contingently — not because retaliation branches exist, but because the damage is baked into the entire distribution the moment the action is taken. Furthermore: every action that can produce a consequence does produce that consequence, in some branch. There is no version of a harmful action in which all its consequences fail to occur somewhere in the distribution. The branch in which you get away with it is real. So is the branch in which you do not. Both are you. This is not a moral warning. It is a structural description of what acting under MWI actually means.

Principle 3 (Non-Tyranny of the Present)**.** The present self holds no privileged authority over the distribution. To foreclose possibilities for future selves — to collapse the options, optimize for one narrow future — is tyranny. It reduces Triple P by delimiting the distributed being.

Principle 4 (Non-Judgment)**.** No branch-self possesses sufficient information to render legitimate judgment on another. The only vantage point from which such judgment is coherent is the complete distributed being — the full spectrum at the end of the waveform. Within any single branch, that depth is structurally unavailable.

Principle 5 (Coherence as Product)**.** Coherence is not chosen. It is the emergent result of increasing Triple P without increasing constraints. When every branch-self has maximum resources to be fully itself, the distributed being is coherent — not through uniformity, but through full expression of every wavelength.

Principle 6 (The Multiplicative Function)**.** Branching multiplies the effects of present actions. Long-term structural investments — habits, relationships, knowledge — propagate across exponentially many downstream branches. The further into the future an action reaches, the greater its multiplicative return.

Principle 7 (The Structural Precondition)**.** QCE does not derive ought from is. It identifies Triple P as the structural ground of any ethical discourse: without probability (existing), possibility (ways of existing), and potential (capacity within existing), there is nothing for ethics to be about. The framework does not say *increase Triple P because it is good*. It says: for any being that has agency, Triple P is what ethics is already about.

Principle 8 (The Quantum Turn)**.** Previous ethical systems were Newtonian: built for a single trajectory through a single world, where the future self is speculative and trust between temporal selves is a gamble. Under MWI, that foundation exists. Every downstream self is real. The present self does not invest in a speculative future; it shapes the inherited conditions of beings that certainly exist. QCE is the ethics native to the physics.

Conjecture (The Eudaimonic Asymptote). Branches that increase Triple P create conditions for further branching, greater complexity, and increased measure within the wave function. Branches that systematically decrease Triple P tend toward reduced measure and eventual extinction of their qualitative lineage. The system evolves over time toward an attractor state in which dominant measure is saturated with flourishing.

This conjecture is not derived here from first principles — the mathematics of branch measure dynamics across cosmological time remains an open problem. It is offered as a structural implication of the framework: if Triple P is what ethics is already about, and if the multiplicative function of branching compounds investments in Triple P forward through exponentially many downstream branches, then the system as a whole has a directionality. Branches that grow flourish and branch further. Branches that do not, wither.

The opposite attractor is the **entropic asymptote**: the condition described in Robert Charles Wilson's *Divided by Infinity* (1998), in which a consciousness persists across branches of diminishing probability, diminishing possibility, and diminishing potential — surviving, but barely, in increasingly degraded conditions approaching the minimum viable branch. The entropic asymptote is the null hypothesis. It is what happens in the absence of Triple P investment. The QCE equation is the structural answer: not divided by infinity, but *multiplied* by it. The present self is not a passive particle being diluted across branches. It is an ancestor planting conditions in every branch simultaneously, conditions that compound forward through the distribution.

The entropic asymptote is the horror of what branching becomes without Triple P investment.

The eudaimonic asymptote is where it goes with it.

The QCE equation is the distance between them.

Chapter 5

Identity, Parfit, and the Degenerate

Our existence is not a deep further fact,
distinct from physical and psychological continuity.

— Derek Parfit, *Reasons and Persons* (1984), p. 281

THE most important disagreement in this paper is with Parfit, and it deserves full treatment — including full respect for what he achieved before we part ways with his conclusion.

Reasons and Persons (1984) is one of the most rigorously argued works in twentieth-century philosophy. Parfit did not approach the problem of personal identity casually. He spent years constructing thought experiments of extraordinary precision — fission cases, teletransportation cases, gradual replacement cases — not to be clever, but because he understood that the question of what we are, over time, is the question that underlies everything else in ethics. If we cannot say what a person is, we cannot say what we owe them, what they owe us, or why the future self the present self is asked to sacrifice for is worth the sacrifice. Parfit saw the depth of the problem more clearly than almost anyone before him, and he refused to look away from the implications.

His conclusion: personal identity cannot handle branching without becoming arbitrary. In fission cases — a person's brain divided, both halves surviving in different bodies — each resulting person has psychological continuity with the original. Neither is determinately the original. Classical identity fails. Parfit's response was not to abandon ethics but to find what actually does the work that identity was supposed to do. His answer: Relation R — psychological connectedness and continuity with the right kind of cause.[1] What matters in survival is not strict identity but the preservation of the psychological relations that constitute a life. This is a serious answer to a serious problem, motivated by genuine moral concern for what we actually owe each other and ourselves.

The disagreement here is not with Parfit's question. It is with his conclusion. When he cannot find a metaphysical fact that picks out which branch is really you, he concludes identity must be abandoned. QCE concludes identity must branch. The difference is one move, but it is the move that changes everything.

This is serious work. Parfit correctly identifies that classical singular identity cannot survive branching without arbitrariness. The error is in the conclusion.

When scientific ontology expands, concepts are not abandoned. They are expanded. The discovery that the universe is vastly larger than the solar system did not make "the universe" incoherent. The discovery that matter is composed of atoms did not dissolve the concept of ordinary objects. The correct response to branching is to understand that a person is a distributed being — that identity, like everything else, scales with the ontology.

> *The distributed being is not a dissolution of identity. It is identity, fully described for the first time, under the correct physics.*

Wittgenstein's contribution here requires care, because it is doing two distinct things that must not be collapsed into one.

The first move: the private language argument (§243 onwards) dissolves the demand for an inner metaphysical identity-fact.[2] Parfit's position implicitly requires that "identity" refer to some privately accessible, intrinsic property that picks out which branch is really you. When he cannot find this fact under MWI, he concludes identity is indeterminate. But Wittgenstein established, from a completely independent direction and decades before Parfit wrote, that the demand for this kind of inner ostensive reference is itself the philosophical picture that has been holding us captive. As §202 states: *"obeying a rule is a practice. And to think one is obeying a rule is not to obey a rule. Hence*

[1] Derek Parfit, *Reasons and Persons* (Oxford: Clarendon Press, 1984), 281.

[2] Ludwig Wittgenstein, *Philosophical Investigations*, §243, trans. G. E. M. Anscombe, P. M. S. Hacker, and Joachim Schulte (Oxford: Wiley-Blackwell, 2009).

it is not possible to obey a rule ‘privately.”’[3] The inner identity-fact Parfit is looking for was never the kind of thing that could do the work he needed it to do. Wittgenstein does not answer the identity question. He dissolves a bad version of it.

The second move belongs to QCE, not to Wittgenstein. Once the bad question is cleared — once we stop looking for a private inner fact that picks out the real branch — the positive criterion becomes available. What makes all branch-selves instances of the same distributed being is not shared psychology, not continuous memory, not convergent values. It is shared causal origin and wave function structure: the same DNA, the same spacetime coordinates, the same causal insertion into the history of the universe, from which the branching begins. That criterion is structural, not private. It is public in the only sense that matters for a distributed being — it is written into the physics of what a person actually is.

Wittgenstein clears the ground. QCE builds on it. These are two distinct moves, and conflating them would leave the argument incomplete.

5.1 The Degenerate

Parfit himself imagined this case. He called the figure the Degenerate: a person whose psychology has changed so radically through a long sequence of individually small steps that the resulting person shares virtually nothing — no memories, no values, no commitments, no characteristic ways of engaging the world — with the person who began the sequence. Parfit used this case to argue that psychological continuity has been broken, and therefore that whatever remains cannot be said to be the same person in any meaningful sense. Identity, he concluded, has dissolved.

QCE draws the opposite conclusion from the same case.

Under deterministic unitary evolution, the wave function did not produce the original person and then, through a sequence of branches, produce a stranger. It produced the distributed being, branching at every quantum juncture, with each

[3] Wittgenstein, *Philosophical Investigations*, §202.

branch inheriting the conditions left by the previous one. The Degenerate is not a different person who happens to share a causal history with the original. The Degenerate is the distributed being, in a branch where the inherited conditions accumulated in a particular direction. Same causal origin. Same wave function structure. Different inheritance.

> *Yes. The burden of proof runs in the opposite direction from where it is usually placed.*

The question is not how QCE defends calling the Degenerate the same distributed being. The question is how anyone who accepts MWI defends calling him a different one. Under deterministic unitary evolution, the wave function does not produce the present self and a stranger who happens to share its history up to a branch point. It produces the distributed being, going every way the physics permits. The branching does not create new persons. It reveals that the person was always a distribution. The Degenerate's psychology diverged because the inherited conditions accumulated that way. Same distributed being. Different inheritance. That is what branching is.

To say the Degenerate is not you because his psychology diverged is to say the light is not the same light because the wavelengths separated. The prism does not produce different lights. It reveals the full structure of the light that was always there.

This grounds Principle 4 — Non-Judgment — precisely. No branch-self can judge another because no branch-self has enough of the wave function in view to make the judgment coherent. Non-Judgment is not moral indifference. It is epistemological honesty about the limits of any partial perspective.

Chapter 6

The Pre-Ethical Claim and Its Objections

In every system of morality which I have hitherto met with,
I have always remark'd that the author proceeds for some time
in the ordinary way of reasoning... when of a sudden
I am surpriz'd to find that instead of the usual copulations
of propositions, is and is not,
I meet with no proposition that is not connected
with an ought, or an ought not.

— David Hume, *A Treatise of Human Nature*, 3.1.1 (1739)

6.1 Against Hume's Guillotine

QCE does not derive ought from is. Hume's is/ought gap is real and correctly identified, and this framework does not cross it. It does something different: it identifies the structural preconditions that must obtain before any normative claim is coherent at all. Without a subject that exists, has ways of existing, and has capacity within existing, no ethical claim has a subject to be about. This is not a value judgment. It is a condition of coherence.

The natural law objection deserves a direct answer: does identifying Triple P as the structural ground of ethics smuggle normativity in under descriptive language, as Aquinas smuggled in teleology with "follow nature"? No. Aquinas derived normative conclusions from alleged purposes embedded in nature — he moved from is to ought via implicit teleology. QCE identifies conditions of predication: Triple P is what must be present for any ethical discourse to have a subject.

Principle 1 follows not from a claim about what nature intends but from the observation that any ethical framework that systematically destroys the conditions of its own subject matter has ceased to be an ethical framework. Whatever good and evil turn out to mean in any normative system, they must mean at least this, or they mean nothing.

> *Triple P is the floor below which the words good and evil lose their referent entirely.*

This position occupies a genuinely novel location in the is/ought landscape — one that Hume's framework does not have a clean category for. It is not a derivation of ought from is. It is the identification of a structural minimum below which ought loses its subject. This is closer to what Kant was doing in the transcendental analytic — identifying conditions of possibility rather than deriving conclusions from nature — except that QCE's conditions are existential rather than epistemic.

6.2 Against the Branch-Weight Problem

The QCE Equation sums branch-weighted Triple P. The Born rule derivation under MWI remains contested.[1] Does QCE depend on a specific resolution?

It does not. Branch weights in the equation serve a descriptive function: they characterize the amplitude structure of the wave function across future branches. But branch weights do not determine the moral weight of any given branch-self. A branch-self with low amplitude is not less of a person than one with high amplitude. It is the complete distributed being, present in a

[1]See David Deutsch, "Quantum Theory of Probability and Decisions," *Proceedings of the Royal Society of London A* 455 (1999): 3129–3137; David Wallace, *The Emergent Multiverse* (Oxford: Oxford University Press, 2012); Wojciech H. Zurek, "Decoherence, Einselection, and the Quantum Origins of the Classical," *Reviews of Modern Physics* 75, no. 3 (2003): 715–775.

less probable branch. The Kantian constraint — never merely as a means[2] — applies to each branch-self fully and without discount, regardless of Born weight. There are no partial persons in the branching structure.

6.3 Against the Motivational Gap

A hardened agent could acknowledge the entire framework and say: yes, I shape the distribution. I simply do not care.

The objection assumes QCE needs to compel action the way deontology compels through duty. It does not. The accurate description already contains everything a rational self-interested agent needs, because of the multiplicative function of branching. Under MWI, a present action propagates through every branch in which it occurs — all downstream branches, because the action is in the causal past of all of them. The investment is multiplicative, not linear.

This closes structurally the oldest gap in the history of ethics. Thrasymachus asked Socrates why the perfectly unjust man who appears just does not simply win.[3] Every prior answer was contingent: reputation, retaliation, divine punishment, the social contract, the categorical imperative. None closes the gap permanently. Under MWI, there is no branch in which the unjust action occurred and all its consequences failed to occur somewhere in the distribution. Thrasymachus was not wrong about the Newtonian universe. He was working with the wrong physics.

> *The question he asked has waited two and a half thousand years for an answer that does not depend on contingency. This is that answer.*

6.4 Against the Locality of Suffering

Suffering is local. The branch-self in pain is in pain here. Why should a distributed ontology bear on moral concern grounded in local experience?

The objection conflates locality of experience with locality of moral concern.

[2]Immanuel Kant, *Groundwork of the Metaphysics of Morals*, Ak. 4:429, trans. Mary Gregor and Jens Timmermann (Cambridge: Cambridge University Press, 2012).

[3]Plato, *Republic*, 338c–339a, trans. G. M. A. Grube, rev. C. D. C. Reeve (Indianapolis: Hackett, 1992).

Only the first follows from the phenomenology. Moral concern has never been coextensive with present experience. The ancestral structure grounds concern for branch-selves by the same logic that grounds concern for any future self: shared causal origin and the inherited conditions that flow from present actions.

The locality of suffering is honored, not dissolved. Non-Judgment does not mean indifference to local suffering. It means epistemic humility about rendering verdict from a vantage point that cannot see the full distribution. The suffering is seen. The judgment is withheld. These are different acts.

Chapter 7

Two Structural Corrections

Soon you'll be ashes, or bones.
A mere name, at most —
and even that is just a sound, an echo.

— Marcus Aurelius, *Meditations*, 5.33
trans. Gregory Hays

PRIOR ethical frameworks were built on a description of the person that was adequate as a practical approximation but incomplete as a metaphysical account. Two corrections follow from the distributed-being ontology.

7.1 Non-Mortality

Non-mortality is not immortality. When a branch-self ends, the distributed being is not exhausted. The distributed being — defined by causal origin and wave function structure — continues in every branch where it continues. The present branch-self is not the whole of what the distributed being is, any more than a single cell is the whole of the organism it belongs to.

The terror that has animated Newtonian ethics — that the end of the branch-self is the end of everything — is a misidentification of the branch-self with the being. Under the correct description, the terror does not disappear as an experience. It loses its metaphysical justification.

7.2 Non-Finitude

Non-finitude is a structural claim, not a size claim. The distributed being is non-finite in the sense of the real numbers between zero and one: bounded, but containing infinitely many points because between any two points, however close, there are infinitely many others. The non-finitude is in the structure. The distributed being cannot be reduced to a finite list of instances because the branching structure that generates those instances is non-finite by its nature.

This is the structural ground of Principle 3 — the Non-Tyranny of the Present. The present self cannot foreclose the distributed being, because the distribution is not the kind of thing that can be foreclosed by any finite action.

These are corrections to a description. The Newtonian description of the person — finite, mortal, bounded, sole instance of itself — was not wrong as a practical approximation. It was wrong as a metaphysical account of what a person actually is under the physics we now have. Every ethical framework built on that description was doing the best philosophy possible with the available ontology. QCE does not replace what those frameworks built. It corrects the description they were building on.

Chapter 8

Conclusion

Non-finitude is your inheritance.
On a long enough timeline,
your is becomes your ought.

— *QCE*, Principle 7

EVERY ethical system ever written was built by someone who did not know they were a distributed being. Aristotle pointed at the ground with eudaimonia. Epicurus correctly diagnosed the root distortion. Marcus Aurelius felt the hive and the bee. Hobbes was honest about what the Newtonian universe requires. Spinoza had the courage to turn the ontology. Hume held the boundary firm. Kant reached for the foundation with both hands and postulated what he could not prove. Rand asked the honest question. Hegel preserved both sides of the branch. The Romantics kept alive the conviction that the mechanical account was missing something essential. Parfit got one move away.

None of them had the physics. The physics arrived in 1957 in a dissertation that the world was not ready to read.

If Everett is right — and the evidence continues to accumulate that he was — then the distributed being is real, the ancestral structure is real, and the ground beneath the ought has been there all along, waiting for the physics to make it visible.

This is not the end of ethics. It is the beginning of ethics on a foundation that actually holds.

Non-finitude is your inheritance.

On a long enough timeline,
your is becomes your ought.

The spectrum is still emerging.

You are not at the end of the light's journey through the prism —

you are inside it, one wavelength,

watching the others diverge.

They are as real as you are.

The question was never who you are.

The question is what you are leaving
in the frequencies you will never inhabit.

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The Oversoul Speculation

The following is offered as speculative philosophical extension,
explicitly not part of the formal framework.
It is not defended by the physics and makes no claims
beyond the conditional and analogical.
It is included because the logic of the distributed being,
followed far enough, opens a question
that deserves to be stated rather than suppressed.

CONSIDER the body. Approximately thirty-seven trillion cells, each alive in some meaningful sense, each carrying the same DNA, the same causal origin. A skin cell on the right toe and a marrow cell in the left femur have never communicated. They do not know each other exists. And yet each one, if asked the impossible question, would answer the same way: *I am this person. Who else would I be?*

The body is the distributed being at the cellular scale. The cells are instances at the biological scale. The person — the unified experiencing subject — is the emergent property of the full distribution, not identical with any single cell but not existing apart from them either.

Now extend the analogy.

The distributed being of QCE — the person refracted across every branch of the wave function sharing their causal origin — is related to its branch-selves the way the body is related to its cells. Each branch-self is a real, experiencing instance of the distributed being. The branch-selves do not communicate across branches, just as cells in distant parts of the body do not communicate directly. And yet each one, experiencing from inside its branch, is fully and completely the distributed being.

The speculative question: if cells are to the person as branch-selves are to the distributed being — is the distributed being itself a cell in something larger?

The Vedantic tradition answered yes and called the larger thing Brahman. Emerson answered yes and called it the Over-Soul: *"That Unity, that Over-soul, within which every man's particular being is contained and made one with all other."*[1] QCE does not answer yes or no. It notes that the logic of the distributed being, applied recursively, does not terminate at the scale of the individual person. Causal origins are embedded in larger causal structures — families, species, biospheres, the causal history of the universe itself. The question the framework opens without resolving: are these larger causal structures themselves distributed beings, with individual persons as their instances?

This is not a claim. It is a question the structure makes available. And it is worth stating, because the history of physics suggests that when a framework opens a question this large, the question is usually worth asking.

The spectrum is most coherent
when no wavelength is suppressed.

You are not making me more me
by cutting away what I could possibly be.

By defining me
you are delimiting me.

[1] Ralph Waldo Emerson, "The Over-Soul," in *Essays: First Series* (Boston: James Munroe, 1841).

Canon Formalization

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Attached so that no reader need take the formalism on faith.

The complete formal apparatus — Foundations 1–5, Triple P definitions, the Canon Equation, dual reading, action selection formalism, eight principles, and glossary — is presented in full in Chapter 4 of this volume. That chapter *is* the canon formalization, reproduced here in its native context rather than as a separate appendix, so that the mathematics and the philosophy inhabit the same space from the beginning.

The separately published canon formalization (DOI above) presents the same apparatus in standalone format for citation in technical contexts.

Term	**Definition**
Distributed being	A single entity refracted across all branches sharing its causal origin
Triple P	Probability, Possibility, Potential — the structural preconditions of ethical discourse
Branch weight $w(\omega\|a)$	Born measure — ontological probability, how much reality flows into a branch
Probability $P(\omega)$	Epistemological — whether you are alive in a given branch
Possibility $\Sigma(\omega)$	Scope — degrees of freedom, open paths, ways existence can be lived
Potential $\Lambda(\omega)$	Scale — capacity, resources, health, knowledge, power to act
Coherence	The emergent state of a distribution in which every branch-self is fully resourced to be itself
Tyranny of the present	Any act by which the present self forecloses possibilities for the distributed being
Newtonian ethics	Any ethical framework built on the assumption of a single timeline and a single future self